

SOP-MNT-010 Pressure Safety Valve Testing and Recertification

Revision C - Owner: Technical Services - Classification: Internal

1. Purpose

This procedure defines how pressure safety valves (PSVs) at KCR are removed, bench tested, overhauled and recertified. A PSV protects equipment from overpressure by relieving fluid when the inlet pressure reaches the set pressure.

2. Test interval

Every PSV shall be bench tested at least once every 24 months. PSVs in fouling, corrosive or polymerising service (for example crude overhead, sour water and amine regenerator service) shall be tested every 12 months. The interval may be extended to 36 months only with a documented risk assessment approved by the Inspection Head.

3. Acceptance criteria

The measured popping pressure shall be within plus or minus 3 percent of the stamped set pressure for set pressures above 5 bar(g), and within plus or minus 0.15 bar for set pressures of 5 bar(g) or less. Seat leakage shall not exceed the limits in STD-ENG-120. A valve that pops outside tolerance on the as-received test shall be recorded as a failed test even if it passes after adjustment.

4. Records

Record the as-received pop pressure, as-left pop pressure, seat tightness result, spring number and the name of the witness. Update the PSV register within 3 working days of the test.

5. Responsibilities

The Maintenance Engineer arranges removal and isolation. The Inspection Engineer witnesses the test. Recertification is signed by the Inspection Engineer, never by the workshop technician alone.